

Final Project Report: Detailed Summary

Name: Shaik Mohammed Arshad

Domain: Python Development / Cybersecurity

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Project Title: SecurePass Password Manager Vault

I. Overview:

The objective of this project was to develop a professional-grade, desktop-based Password Manager designed to securely store and organize user credentials. The system is engineered to solve common security risks like weak password reuse by providing a centralized, locally-stored encrypted vault.

II. Technical Implementation & Achievements:

1. Advanced Security (Argon2id):

- Implemented the Argon2id hashing algorithm, which is globally recognized for its memory-hardness and resistance to GPU-based brute-force attacks.
- Developed a robust master-authentication gate that protects the entire database, ensuring that even if the database file is compromised, the contents remain encrypted.

2. Relational Database Design (SQLite3):

- Designed a relational schema to map service names to their respective usernames and securely hashed passwords.
- Optimized data retrieval through SQL 'LIKE' queries, enabling a fast and case-insensitive search feature.

3. User Interface Engineering (Tkinter):

- Built a multi-screen application using the Tkinter framework, featuring a separate secure login phase and an internal management dashboard.
- Integrated event-driven functions to ensure that all UI actions—such as adding a password or searching the vault—interact directly with the secure backend logic.

III. Challenges & Solutions:

- **Parameter Tuning:** Finding the optimal balance between Argon2's time and memory costs to ensure security without causing lag on lower-spec hardware.
- **UI State Management:** Effectively clearing and regenerating the application window during screen transitions to prevent memory leaks or residual UI elements from appearing.

IV. Learning Resources:

- Consulted official library documentation (argon2-cffi, sqlite3, tkinter) for best practices in secure coding.
- Followed industry-standard cybersecurity tutorials to understand current threats and mitigation strategies.

V. Final Status & Growth:

The Password Manager is now a fully functional desktop tool that meets all original project requirements. The project demonstrates a successful bridge between complex backend security algorithms and a functional frontend user experience.